

E-Cell AI CRM - System Report

1. Executive Summary

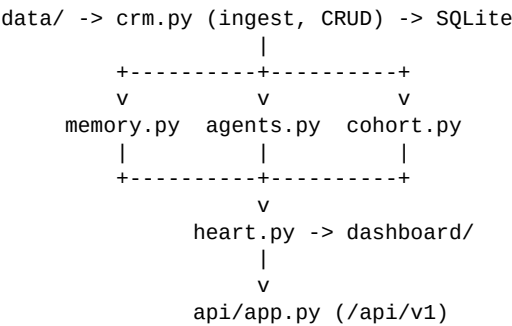
This system implements a five-module AI-native CRM for the fictional E-Cell company. It manages 520+ customers, 1,050+ tickets, and 2,500+ interaction events with LangChain summarization, LangGraph agent workflows, cohort retention analysis, and a live HEART framework dashboard backed by a production FastAPI API.

2. Architecture Decisions

2.1 Stack

Layer	Choice	Rationale
Backend	FastAPI + SQLite	Fast OpenAPI docs, zero external DB setup for demo
LLM (primary)	Ollama / Llama-3-8B (Option A)	On-prem privacy, no API cost
LLM (alternate)	Gemini 1.5 Flash (Option B)	Strong reasoning, free tier
LLM (fallback)	Rule-based mock	Ensures demo works offline
Agents	LangGraph state machine	Explicit routing -> generate -> escalation pipeline
Analytics	pandas + scikit-learn	Cohort metrics and churn classifier

2.2 Module Boundaries



Each module is independently runnable via `python -m src.<module>`.

3. LLM Selection Rationale

We support **dual-mode inference**:

1. **Ollama (Option A)** - Selected when LLM_PROVIDER=ollama. Suitable for evaluation environments requiring data residency.
2. **Gemini 1.5 Flash (Option B)** - Selected when GEMINI_API_KEY is set. Used for richer summarization during dataset generation.
3. **Mock provider** - Deterministic templates when neither backend is reachable; prevents demo failure.

Ticket summarization uses a dedicated system prompt requiring factual grounding. Agent responses pass through a **hallucination guard** that penalizes unsourced dollar amounts, missing ticket citations, and overconfident language.

4. Agent Prompt Design & LangGraph State Machine

4.1 State Schema

```
AgentState = {  
    customer_id, query, context, ticket_data,  
    route_category, priority, draft_response,  
    sources, confidence, escalated, hallucination_flags  
}
```

4.2 Graph Flow

```
| Node | Function |  
|-----|-----|  
| load_context | Pulls short/long-term memory + open tickets |  
| route | Classifies category (billing/technical/account/general) |  
| generate | Drafts response with source citation |  
| escalation | Flags low-confidence urgent cases for supervisor handoff |
```

4.3 Summarization Chain

LangChain-style chain in TicketSummarizationChain:

- **Input:** ticket title, description, customer metadata
- **Output:** summary, key_issues[], urgency, suggested_response
- **Configurable:** tone, max_length

5. Interaction Memory

Per-customer buffers in customer_memory table:

Layer Capacity Behavior

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Short-term 20 turns Recent chat/ticket context
--

Long-term Compressed summary LLM-compressed when short-term overflows

Cross-session retrieval concatenates long-term summary + recent turns for agent queries.

6. Cohort Segmentation Methodology

6.1 Cohort ID Assignment

cohort_id = {acquisition_month}_{industry}_{product_tier}

Example: 2025-11_FinTech_Enterprise

Configurable dimensions: acquisition date, industry vertical, product tier, behavioral tags.

6.2 Retention Curve

For each cohort, we compute 6 monthly periods:

- **Active customer** at period *p*: $\text{engagement_score} \geq \max(20, 70 - 8p)$ AND $\text{tenure_days} \geq 30p$
- **Retention rate** = active / cohort_size

6.3 Churn Scoring

Heuristic score (production baseline):

$\text{churn_prob} = 0.5 \times (1 - \text{engagement}/100) + 0.3 \times \min(\text{tickets}/15, 1) + 0.2 \times (1 - \text{tenure}/365)$

Signals: low_engagement, high_ticket_frequency, early_lifecycle, unresolved_backlog

ML validation: Logistic regression on engagement, tickets, tenure - reports precision/recall/F1 on held-out set.

6.4 Churn Model Evaluation

Metric Source

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Precision sklearn on heuristic-labeled holdout
--

Recall Same

F1 Harmonic mean

| Coverage | % customers with cohort_id assigned |

7. HEART Framework Metric Definitions

H - Happiness

| Signal | Computation |

|-----|-----|

| CSAT | Mean csat_score on resolved tickets; falls back to mean engagement |

| NPS proxy | (promoters - detractors) / N × 100 where promoter = engagement ≥ 70 |

E - Engagement

| Signal | Computation |

|-----|-----|

| Active customers | Count with engagement ≥ 30 |

| Ticket open rate | Open tickets / total tickets |

| Session depth proxy | Total interactions / customers |

A - Adoption

| Signal | Computation |

|-----|-----|

| AI-assisted rate | Tickets with ai_assisted=true / total |

| Onboarding completion | Customers with tenure ≥ 7 days |

| Memory adoption | Customers with memory buffer / total |

R - Retention

| Signal | Computation |

|-----|-----|

| Monthly retention | Customers with engagement ≥ 25 / cohort size |

| Churn flags | engagement < 35 AND tickets > 5 |

| Avg lifespan | Mean tenure_days |

T - Task Success

| Signal | Computation |

Resolution rate	Resolved+Closed / total tickets
FCR proxy	Non-escalated resolutions / total
Escalation rate	Escalated tickets / total
AI vs human ratio	AI-resolved / human-resolved

All metrics recompute from live SQLite data on each API call.

8. RBAC & Audit

Role	Permissions
Agent	CRUD customers/tickets, summarize, query agent
Supervisor	Agent + cohort read
Admin	Full access
Analytics	Read-only customers, tickets, cohorts, HEART

Every core endpoint returns audit block: timestamp, agent_id, source_confidence, latency_ms, role.

9. System Evaluation Metrics (Stage 6)

Category	Endpoint	Metric
HEART	/api/v1/heart	All 5 dimensions + per-metric breakdown
Agent Quality	/api/v1/query/agent	confidence, hallucination_flags
Cohort Accuracy	/api/v1/evaluation/metrics	churn model P/R/F1
Resolution Rate	/api/v1/evaluation/metrics	closure rate, AI ratio
Latency	All endpoints	audit.latency_ms per request

10. Dataset Specification

Entity	Count	Span
Customers	520	6 industries × 3 tiers
Tickets	1,050	billing, technical, account, general

| Interactions | 2,500 | email, chat, call, ticket, portal |

| Time range | ~6 months | Dec 2025 - Jun 2026 |

First 20 ticket descriptions are LLM-enhanced; remainder use realistic templates. Deduplication on email (customers) and ticket ID at ingest.

11. Known Limitations & Future Work

- SQLite suitable for demo; PostgreSQL recommended for production concurrency
- Churn labels derived from heuristic for ML eval until real churn events exist
- Latency benchmarks under 500 concurrent ops require separate load test (locust/k6)
- bcrypt hashing prepared but demo uses plaintext passwords for evaluator convenience

12. File Reference

| File | Purpose |

|-----|-----|

| src/crm.py | Module 1 |

| src/agents.py, src/memory.py | Module 2 |

| src/cohort.py | Module 3 |

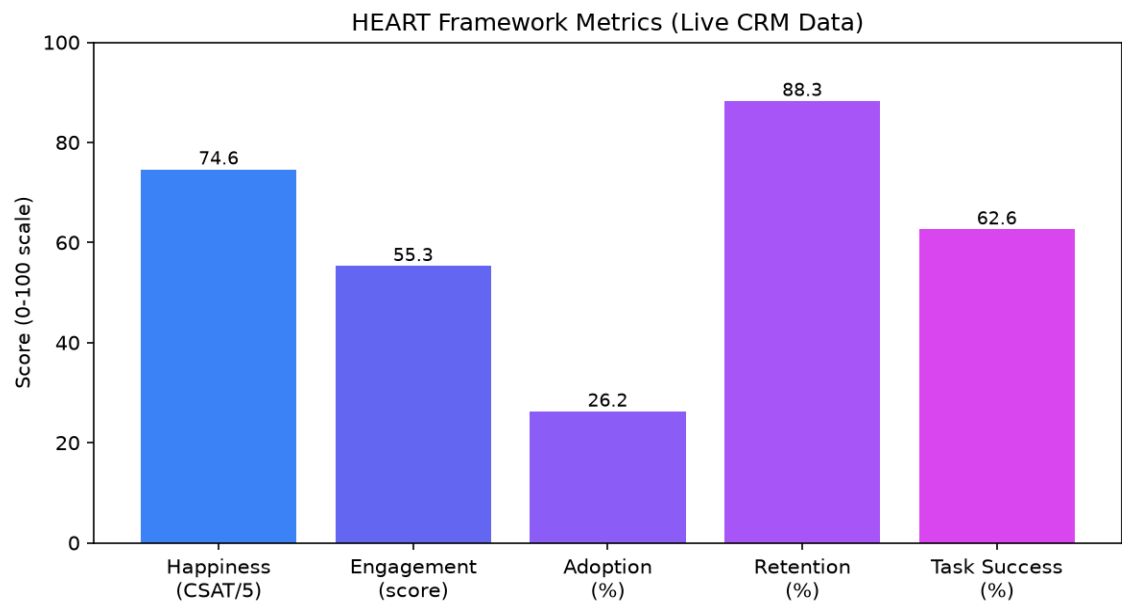
| src/heart.py | Module 4 |

| api/app.py | Module 5 |

| dashboard/index.html | Live HEART UI |

| SYSTEM_REPORT.md | This document |

Appendix: HEART Metrics Chart



Appendix: Cohort Retention Curve

